

who are too short suffer because they cannot excel at pickup basketball games, but some sweet spot in the middle of too tall and too short tends to maximize happiness. Even if we can express this function perfectly and apply gradient ascent to find the maximum, it will not be useful to us, because we do not have control over our height.

If we zoom out even further, we may have all the ingredients required for gradient ascent (or any other algorithm) and still wish to refrain for deeper philosophical reasons. For example, suppose you can precisely determine a tax-revenue function and you're elected prime minister with full control over the taxation rate in your country. Before you apply gradient ascent and climb to the revenue-maximizing peak, you may want to ask yourself if maximizing your tax revenue is the right goal to pursue in the first place. It could be that you are more concerned with freedom or economic dynamism or redistributive justice or even opinion polls than you are with state revenues. Even if you have decided that you want to maximize revenues, it's not clear that maximizing revenues in the short term (that is, this year) will lead to maximization of revenues in the long term.

Algorithms are powerful for practical purposes, enabling us to achieve goals like catching baseballs and finding revenue-maximizing taxation rates. But though algorithms can achieve goals effectively, they're not as suited to the more philosophical task of deciding which goals are worth pursuing in the first place. Algorithms can make us clever, but they cannot make us wise. It's important to remember that the great power of algorithms is useless or even harmful if it is used for the wrong ends.

Summary

This chapter introduced gradient ascent and gradient descent as simple and powerful algorithms used to find the maxima and minima of functions, respectively. We also talked about the serious potential problem of local extrema, and some philosophical considerations about when to use algorithms and when to gracefully refrain.

Hang on tight, because in the next chapter we discuss a variety of searching and sorting algorithms. Searching and sorting are fundamental